

UE23CS320A - Capstone Project Approval



Project Title :Drishti: AI-Powered Early Crisis Prediction and Advisory System.

Project ID :80

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DRISHTI: AI-POWERED EARLY CRISIS PREDICTION AND ADVISORY SYSTEM



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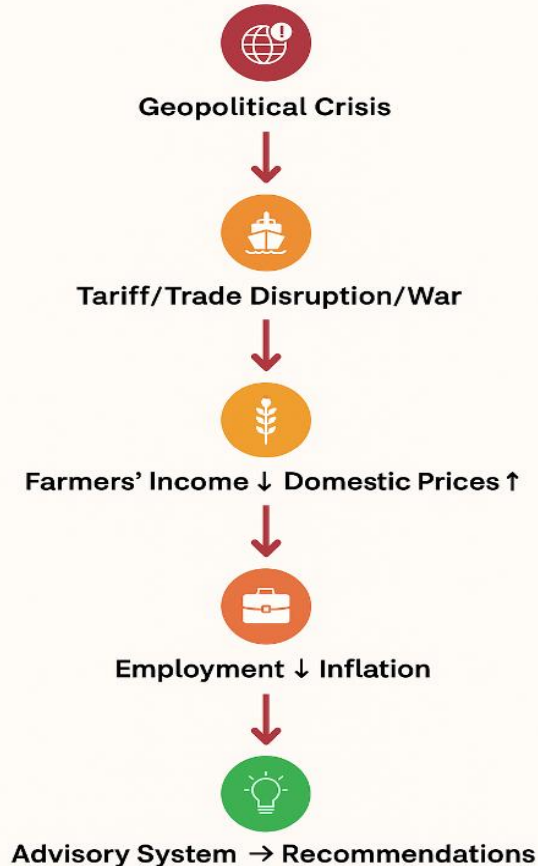
—TIMELINE

Drishti: AI-Powered Early Crisis Prediction and Advisory System

Problem statement:

- ❖ Global geopolitical crises such as tariffs, policy changes, conflicts, and trade disruptions – have a direct and severe impact on the Indian economy, particularly agriculture. These shocks affect exports, GDP, employment, forex, and inflation, but their impacts are usually understood reactively—only after losses occur.
- ❖ Gap in Current Solutions:
While existing AI solutions focus on crop yield or price forecasting, they do not capture the impact of global geopolitical crises on agriculture and the wider economy.
- ❖ **Drishti** fills this gap by providing an AI-powered early warning and **advisory system**, starting with Indian agriculture and scalable to other key sectors.

PROPOSED SOLUTION: (with Cascading Effects)



- **Data Collection:** Export stats + crisis events (News APIs, GDELT).
- **Cascading Impact:** Crisis → Trade disruption → Income ↓ → Prices ↑ → Employment ↓ → Inflation ↑ → GDP/Forex strain.
- **Prediction & Advisory:** AI/ML models forecast impacts & provide actionable recommendations for farmers, govt., policymakers.
- **Scalability:** Extendable beyond agriculture (energy, healthcare, manufacturing).

Domain: Impact analysis of global geopolitical events on Indian sectors (starting with agriculture).

Focus: Early warning, forecasting, and advisory system.

Coverage:

Agriculture (exports, yields, pricing, farmer income, supply chain).

Ripple effects on GDP, employment, forex, inflation.

Scalability: Extendable to other sectors (e.g., textiles, IT, manufacturing).

Stakeholders: Farmers, policymakers, trade bodies, financial institutions.



FEASIBILITY STUDY

ECONOMIC FEASIBILITY

Early warnings reduce losses from trade shocks and price volatility.

Cost-effective compared to reactive crisis management.

Potential high ROI for stakeholders.

TECHNICAL FEASIBILITY

Use of ML/AI methods (time-series, NLP, sentiment analysis).

Availability of datasets (trade data, WTO/FAO reports, news feeds).

Cloud platforms ensure scalability and real-time processing.

SOCIAL FEASIBILITY

Empowers farmers with timely advisories.

Supports policymakers in proactive planning.

Promotes food security, market stability, and public trust.

OPERATIONAL FEASIBILITY

Accessible via web/mobile dashboards.

Real-time alerts for farmers, policymakers, and institutions.

Integrates smoothly into existing decision-making workflows

POSSIBLE SHORTCOMINGS/CHALLENGES

SHORTCOMINGS OF EXISTING WORK

Focus mainly on crop yield/price prediction.

No linkage of geopolitical events → economic impacts.

Limited tools for real-time early warning/advisory.

CHALLENGES FOR PROPOSED WORK

Data Gaps: Incomplete/irregular data on crises & trade.

Uncertainty: Geopolitical events are sudden & unpredictable.

Model Complexity: Integration of time-series, NLP, sentiment analysis.

Interpretability: Need to simplify outputs for non-technical users.

Scalability: Expanding from agriculture to other sectors.

Use Cases

A stylized illustration of a woman with short dark hair, wearing a black top and blue pants, pointing her right hand towards a large light blue rectangular area on the left side of the slide. She is holding a blue folder or tablet under her left arm.

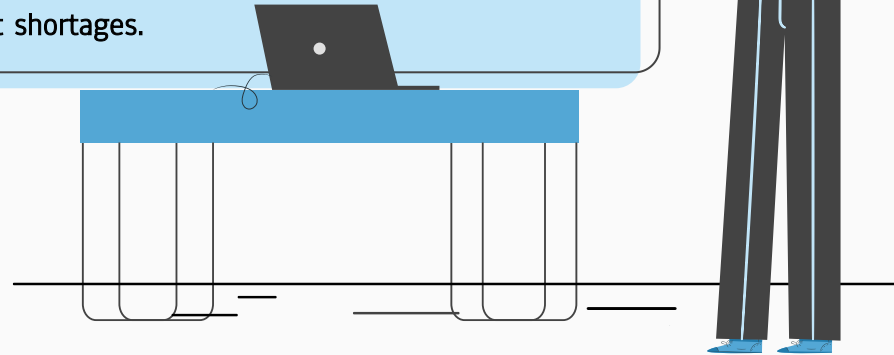
Tariff Shock – U.S. imposes 50% tariff on rice → exports drop → Govt. supports farmers & shifts to Africa markets.

Import Ban – If EU bans Indian mangoes, local prices crash → Exporters quickly sell to Gulf or Asian markets instead.

Conflict Impact – If a war (like Russia–Ukraine) raises fertilizer prices, the model predicts food inflation → Govt. increases subsidy on fertilizer to help farmers.

Policy Shift – If Europe introduces carbon rules on sugar imports, the model shows exports may fall → Exporters adopt cleaner production to meet rules.

- Policy Planning – The government can know in advance how a global event (like a tariff or ban) will affect Indian crops and prepare early.
- Export Councils – APEDA/FIEO identify risk crops & new markets.
- Farmer Support – Early warnings help stabilize incomes.
- Agri-Business – Exporters/processors adjust pricing & sourcing.
- Finance/Insurance – Banks assess agri-loan & crop risk.
- Supply Chain – Predict fertilizer/input shortages.



Background work

Sl. No	Title of the publication with Publisher	Authors/Year	Models used	Key Idea	Limitations
1.	NEWS EVENT-DRIVEN FORECASTING OF COMMODITY PRICES	Chakraborty, Subramanian 2024	● K-means clustering, ● REN (Recurrent Event-driven Neural Network, RNN-based), ● CRF (Conditional Random Fields)	They combine news text + historical price data to forecast agricultural commodity prices. The novelty is in extracting structured event embeddings from news (like “policy change,” “fuel hike,” “elections”) and feeding them into an RNN to improve commodity price prediction.	● Focused mainly on price forecasting, not on broader economic or agricultural impact assessment. ● Works for commodities but does not extend into a policy advisory system
2	MACHINE LEARNING-DRIVEN GOVERNANCE: PREDICTING THE EFFECTIVENESS OF INTERNATIONAL TRADE POLICIES THROUGH POLICY AND GOVERNANCE ANALYTICS.	Emran Hossain 2024	● Regression/Time Series: Linear Regression, Random Forest ● Boosting, ARIMA, LSTM ● Classification: Logistic Regression, fSVM ● NLP: Bag-of-Words, Transformers for policy documents	Machine learning + NLP to forecast the effectiveness of international trade policies (tariffs, trade agreements, non-tariff barriers). It demonstrates that ML models (esp. Random Forest, LSTM) outperform traditional econometrics in predicting trade outcomes, with policy sentiment signals further improving accuracy.	● Black-box nature of deep learning models (limited explainability). ● Bias if data underrepresents certain countries/sectors.

Sl. No	Title of the publication with Publisher	Authors/Year	Models used	Key Idea	Limitations
1.	A MACHINE LEARNING APPROACH OF EXPLORING NEXUS BETWEEN GEOPOLITICAL RISK AND INDIAN ECONOMIC UNCERTAINTY: A COMPARATIVE REVIEW	Bharat Kumar Meher Abhishek Anand 2025	● RMSE(Validation) ● MSE(Validation) ● MAE(Validation) Calculate the R-square value for each model and select a particular model based on the value.	Economic uncertainties were predicted using geopolitical risk (GPR) and two Indian uncertainty indices (old and new), and predictions from the best-performing models were analysed to understand how geopolitical risks drive Indian economic uncertainty.	● Model limits & assumptions in MATLAB ● Data's time period restrictions ● Unaccounted external economic factors ● Limited generalizability beyond India.
2	THE EFFECTS OF GEOPOLITICAL UNCERTAINTY IN FORECASTING FINANCIAL MARKETS: A MACHINE LEARNING APPROACH	Vasilios Plakandaras Periklis Gogas Theophilos Papadimitriou 2018	● Support Vector Regression (SVR) ● Ordinary Least Squares (OLS) Regression ● Random Walk (RW) Model	The paper gives us idea about geopolitical risk in 14 developing countries can predict oil prices,exchange rates,gold prices using ML algorithms.	● Limited to emerging economies' geopolitical risk. ● Forecast horizon effectiveness fades beyond year ● Monthly data misses short-term shocks. ● GPR index construction may overlook events.

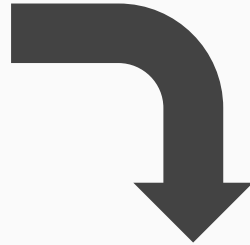
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5.	The Effects of Geopolitical Uncertainty in Forecasting Financial Markets: A Machine Learning Approach	Vasilios Plakandaras et al., 2019	● Support Vector Regression (SVR), ● compared with OLS & Random Walk	Uses Geopolitical Risk Index + financial market data in Support Vector Regression to test predictability of oil, FX, and gold; novelty is finding that gold acts as a consistent “safe haven” while other assets are less predictable.	● SVR often fails to outperform Random Walk; GPR effects are regional/transitory; ● lacks high-frequency & sentiment integration.
6.	Predictive Modeling of US Stock Market and Commodities: Impact of Economic Indicators and Geopolitical Events Using ML	Md Tushar Khan et al., 2024	● Linear Regression ● Random Forest ● XGBoost	Integrates economic indicators, commodity/stock price history, and geopolitical events into ML models (RF, XGBoost) to capture nonlinear market dynamics; novelty is very high accuracy ($R^2 \approx 0.99$) vs. traditional baselines.	● Over-reliance on historical data; poor performance under sudden shocks (wars, pandemics); ● black-box models; ● limited interpretability

Sl. No	Title of the publication with Publisher	Authors/Year	Models used	Key Idea	Limitations
7.	International Trade Economic Analysis for Indian Agriculture during WTO Regime, EJAFR	M. B. Dastagiri et al. 2021	Compound Annual Growth Rate (CAGR) Instability Index (Coefficient of Variation) Export–Import Price Elasticities Regression Analysis (Ordinary Least Squares – OLS)	Studied India’s agri-trade (1990–2016) using growth, volatility, and regression models under WTO. Found comparative advantage in many crops, but price instability often hurt farmers and poverty — except oilseeds, which showed positive impact.	Retrospective only; limited to WTO regime and historical prices no prediction of future geopolitical shocks or cascading effects.
8.	Comparative Analysis of Machine Learning Algorithms to Forecast Indian Stock Market, ITM Web of Conferences (ICDSAC 2023)	G. Naga Chandrika et al. 2023	Generalized Linear Model (GLM) Decision Tree (DT) Random Forest (RF) Gradient Boosted Trees (GBT) Support Vector Machine (SVM) Linear Regression (LR)	Benchmarked multiple ML algorithms on BSE index data (2017–2023) and found that Linear Regression provided the most accurate forecasts, proving ML’s effectiveness in capturing stock market trends.	Focus only on stock data; ignores external/geopolitical factors; limited to financial domain

EXPECTED DELIVERABLES

PHASE - I DELIVERABLES

- The **problem statement document** clearly defines *Drishti* as addressing the impact of global crises on agriculture.
- The system design has 3 layers:
 1. Input layer
 2. Processing layer
 3. Output layer.

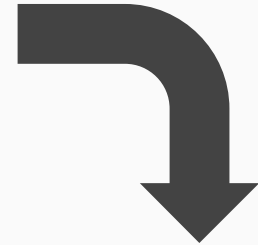
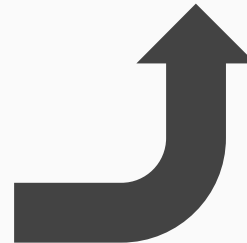


PHASE - II DELIVERABLES

1. Extended Literature survey.
2. Project Requirement Specifications
3. Design philosophy.
4. Baseline Models.

PHASE-III DELIVERABLES

- LLD Blueprint
- Technology Stack Finalization
- Module Development
- Validation Plan
- System Demonstration
- Comprehensive Report



PHASE-IV DELIVERABLES

Publish a research paper in a **Scopus-indexed** or **Web of Science-indexed** journal or conference.

PROJECT TIME LINE

1

- Literature survey on global crises & agriculture.
- Define scope, stakeholders, and deliverables
- Design high-level system architecture.

2

- Collect crisis, trade & agriculture datasets.
- Preprocess data & engineer features.
- Build baseline ML models for impact prediction

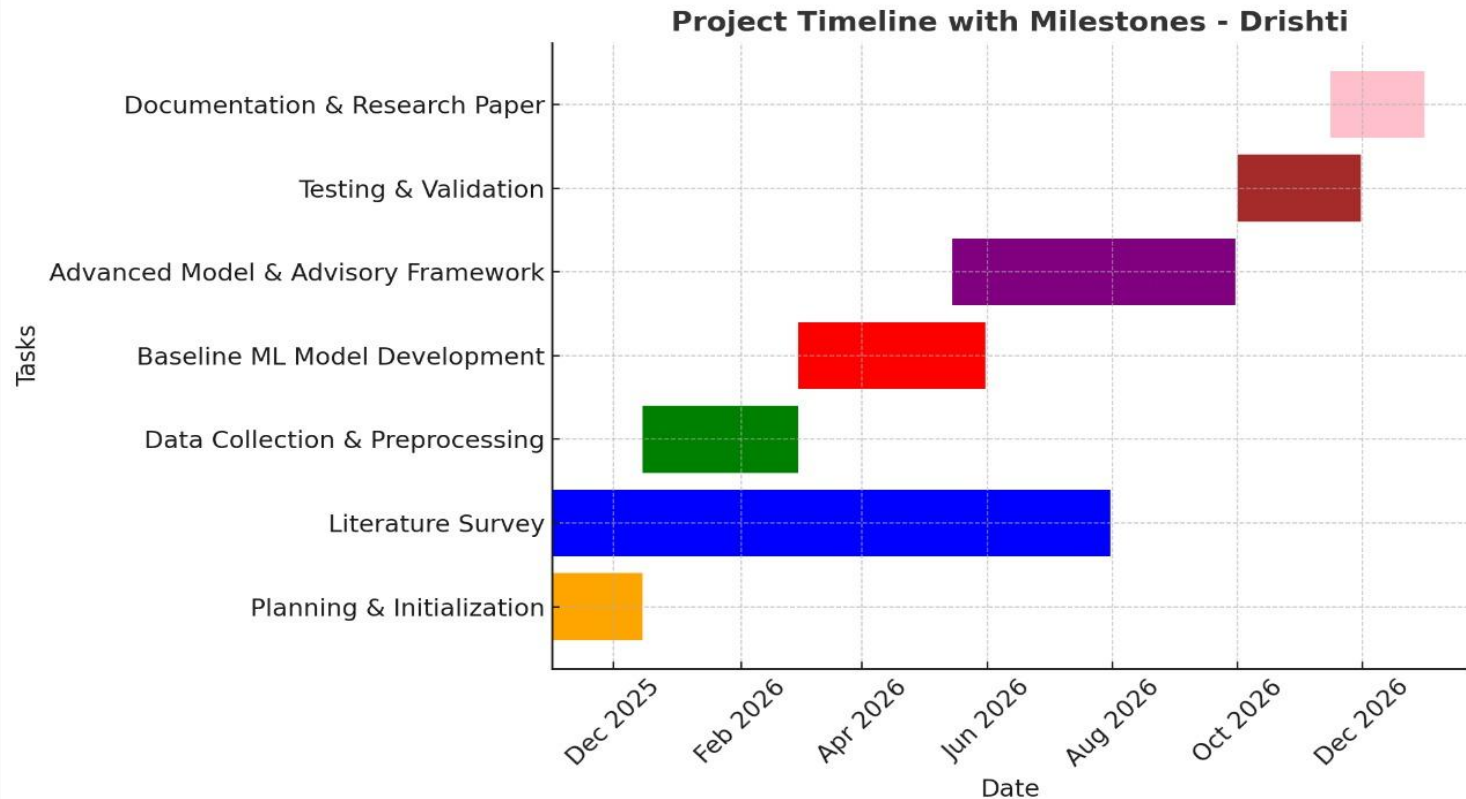
3

- Develop advanced forecasting models.
- Add scenario-based “what-if” simulations.
- Build advisory framework with sectoral alerts

4

- Run final experiments & validate outputs.
- Refine advisory system for usability.
- Write report, publish findings, and propose future work

GANTT CHART



***THANK
YOU***

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